

HAT-P-37b

R-2590

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_150816 · created 2026-08-27T23:14:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457250.6711 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.046
Transit depth (Rp/Rs)^2	2.68 +/- 1.5 [%]
Orbital Inclination (inc)	84.38 +/- 2.24
Airmass coefficient 1 (a1)	1.637 +/- 0.1
Airmass coefficient 2 (a2)	-0.45 +/- 0.44
Scatter in the residuals of the lightcurve fit is	6.09 %
Best Comparison Star	#3 - [152, 184]
Optimal Aperture	2.59
Optimal Annulus	9.01
Transit Duration (day)	0.052 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.164 $\pm$ 0.046 vs 0.1394 (deviation 0.35 σ)
Duration fit vs published	0.052 d vs 0.0983 d (deviation 1.13 σ)
Mid-time O-C	-2.5 min
Depth SNR	2.3
Residual scatter	6.09 %

Light curve

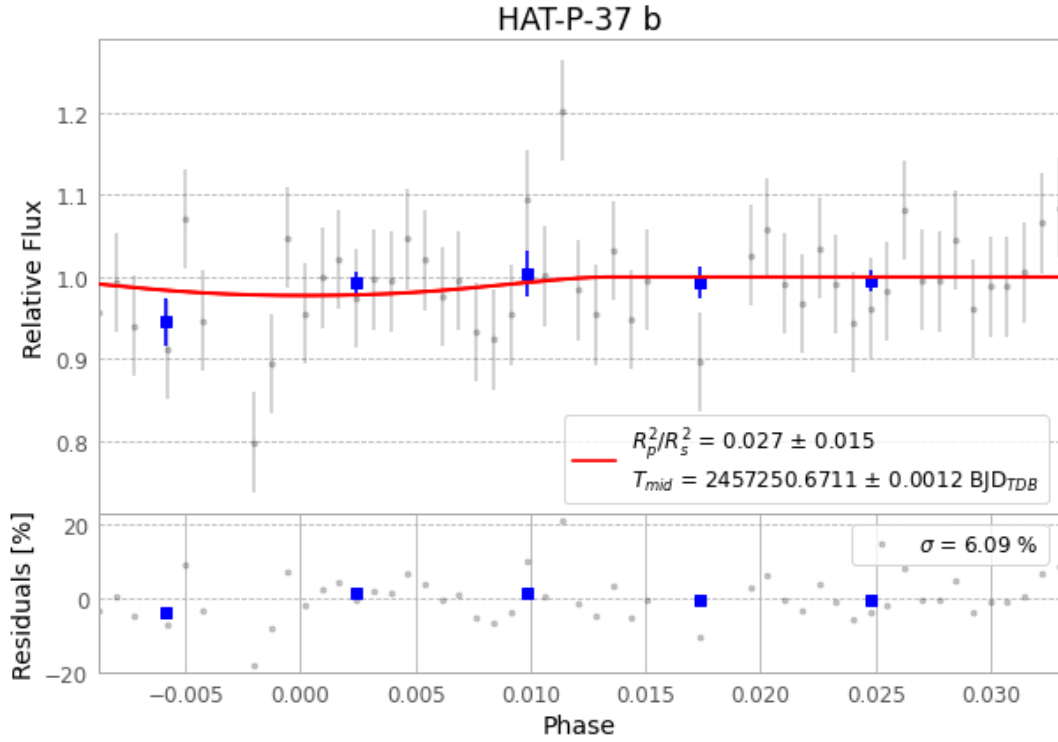


Figure 1 — FinalLightCurve_HAT-P-37 b_2015-08-15.png (SHA-256 40f32958f93f904f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [158, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 622800323623935e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), HATP-37150816032712.FITS → HATP-37150816061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-150816.log (bf01ae01950e...), FinalLightCurve_HAT-P-37 b_2015-08-15.png (40f32958f93f...), FinalLightCurve_HAT-P-37 b_2015-08-15.csv (e9726c701d62...)

Compute resources

Wall time 128.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:14Z.